

REPORT FRONT SHEET
MOLECULAR BIOLOGY, Master Course
EXPERIMENTAL CELL BIOLOGY

NAME OF EXPERIMENT: Exercise 3 week 1: Transfection and Luciferase

DATE FOR PERFORMANCE OF EXPERIMENT: 15-06-2026 -- 26-06-2026

TEACHER: Stine Bull Jessen

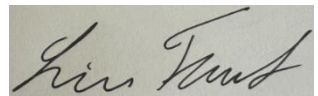
TEAM Green GROUP NUMBER: 3

Names and signatures

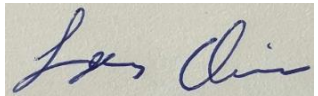
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DATE(S) FOR HANDING IN THE REPORT TO THE TEACHER:

Date for 1. Time
26.06.2026

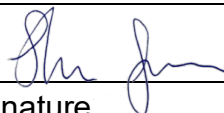
Date for 2. time

Date for 3. Time

TEACHER COMMENTS:

Approved

REPORT APPROVED BY TEACHER:


Signature

26.06.26

Date

This is the front page for the reports and has to be attached to the report. The original front page should be attached when a revised report is handed in.

THE BIOLOGY STUDY BOARD

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Exercise 3 Transfection and Luciferase

Introduction:

Transfecting cells with transgenes can be used for a variety of purposes such as producing deletions, probing tissue specific promoters or equipping cells with fluorescent markers for e.g. lineage tracing.

In this experiment, CRISPR is used for chromosomal integration of a bioluminescence gene, Lucia, in Caco-2 and DLD-1 cells. Depending on how the gene is transfected, the readings gathered will give different results. To test this, three cell conditions are created: Control, CRISPR and Mock. In the control condition, the cell is transfected with a luciferase reporter plasmid, which results in the Lucia gene slowly being phased out as the cell proliferates, meaning the luciferase activity remains short term. The CRISPR condition transfects the genome of the cell through CRISPR mediated gene editing. This leads to a more stable expression of the gene with the lucia gene being passed down when the cell proliferates. Finally, the Mock condition contains no plasmid and serves as baseline control. It measures background signals during the luciferase assay and ensures that successful luciferase activity is detected from the transfection.

Methods:

DLD-1 and Caco-2 cells were treated as in figure 1. Briefly, cells were seeded in 6 well plates and then with treated either pUC18 (mock), pSelect-Zeo-Lucia-AAVS1 (control) or pSelect-Zeo-Lucia-AAVS1+ AAVS1-T2-CRISPR in pX330 (crispr) using polyethyleneimine for transfection. Media was changed the day after, and then two days after media was extracted and then light emission was quantified on a GloMax Luminometer.

Additionally, for Caco-2, control and CRISPR treated cells were seeded in 24 well plates and then 20 μ l media was extracted for a proliferation assay to quantify the amount of luciferase in the media over 4 days.

The workflow for the transfection and luciferase assay, and the luciferase proliferation assay is displayed on figure 1 below, as a schematic presentation.

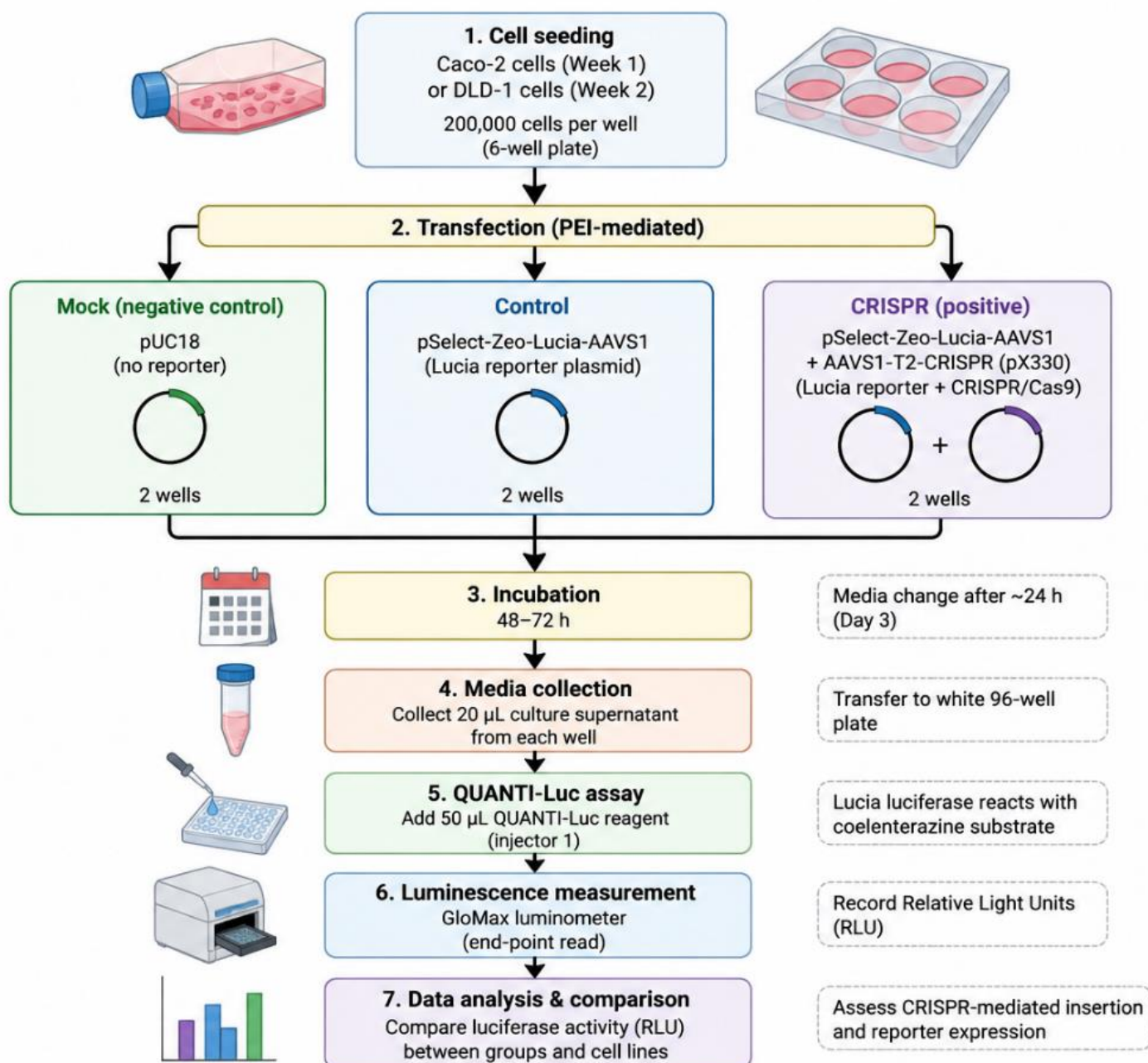


Figure 1. Experimental workflow for CRISPR-mediated luciferase reporter assay. Caco-2 cells (week 1) and DLD-1 cells (week 2) were seeded and transfected using polyethyleneimine (PEI). Three experimental conditions were included: mock transfection (pUC18 plasmid), control transfection (pSelect-Zeo-Lucia-AAVS1 reporter plasmid), and CRISPR-mediated transfection (pSelect-Zeo-Lucia-AAVS1 together with AAVS1-T2-CRISPR/Cas9 plasmid). Following incubation, culture supernatants were collected and Lucia luciferase activity was measured using the QUANTI-Luc assay and a GloMax luminometer. Luciferase activity was recorded as relative light units (RLU) and compared between transfection conditions and cell lines to evaluate CRISPR-mediated reporter integration and expression. Schematic presentation of workflow was generated with help from AI.

Results:

Results from Transfection and Luciferase assay

Results from transfection and luciferase assay is displayed on figure 2 below for the Caco-2-cell line. Results from the assay indicated that cells transfected with the luciferase reporter plasmid displayed substantially higher luciferase activity than mock-transfected cells. The CRISPR-transfected cells showed luciferase activity comparable to or higher than the control group, indicating successful reporter expression following CRISPR-mediated integration. In contrast, mock-transfected cells exhibited only low background luminescence, confirming that the measured signal originated from Lucia reporter expression. Standard deviations (std) from CRISPR-transfected cells had higher variance than std from control cells.

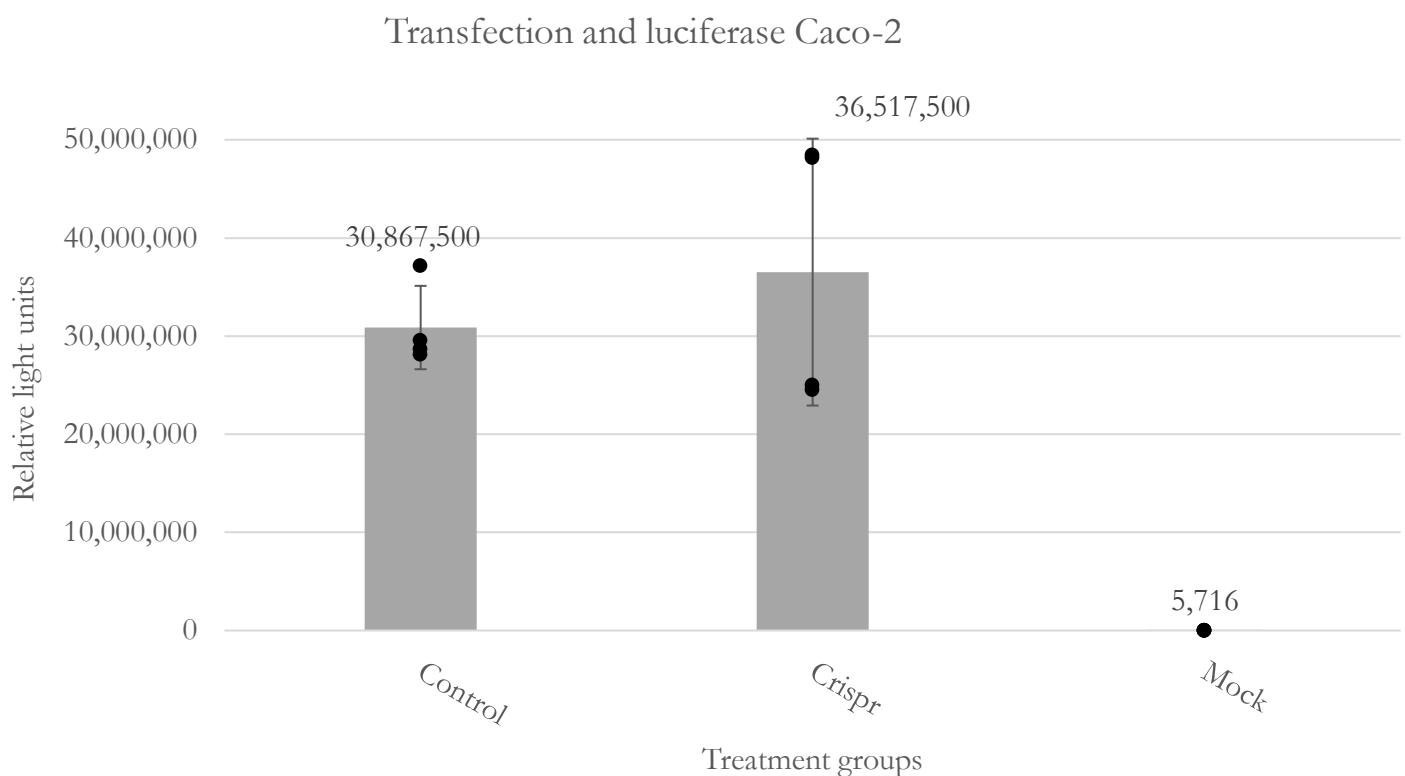


Figure 2. Luciferase activity following transfection of Caco-2. Cells were transfected with pUC18 (Mock), pSelect-Zeo-Lucia-AAVS1 (control), or pSelect-Zeo-Lucia-AAVS1 together with AAVS1-T2-CRISPR (CRISPR). Luciferase activity was measured in culture supernatants using the QUANTI-Luc assay and expressed as relative light units (RLU). Bars represent mean $\pm$ SD. Made with excel.

Figure 3 below display that DLD-1 cells demonstrated strong luciferase activity in the reporter-containing groups, whereas mock-transfected cells showed only minimal background signal.

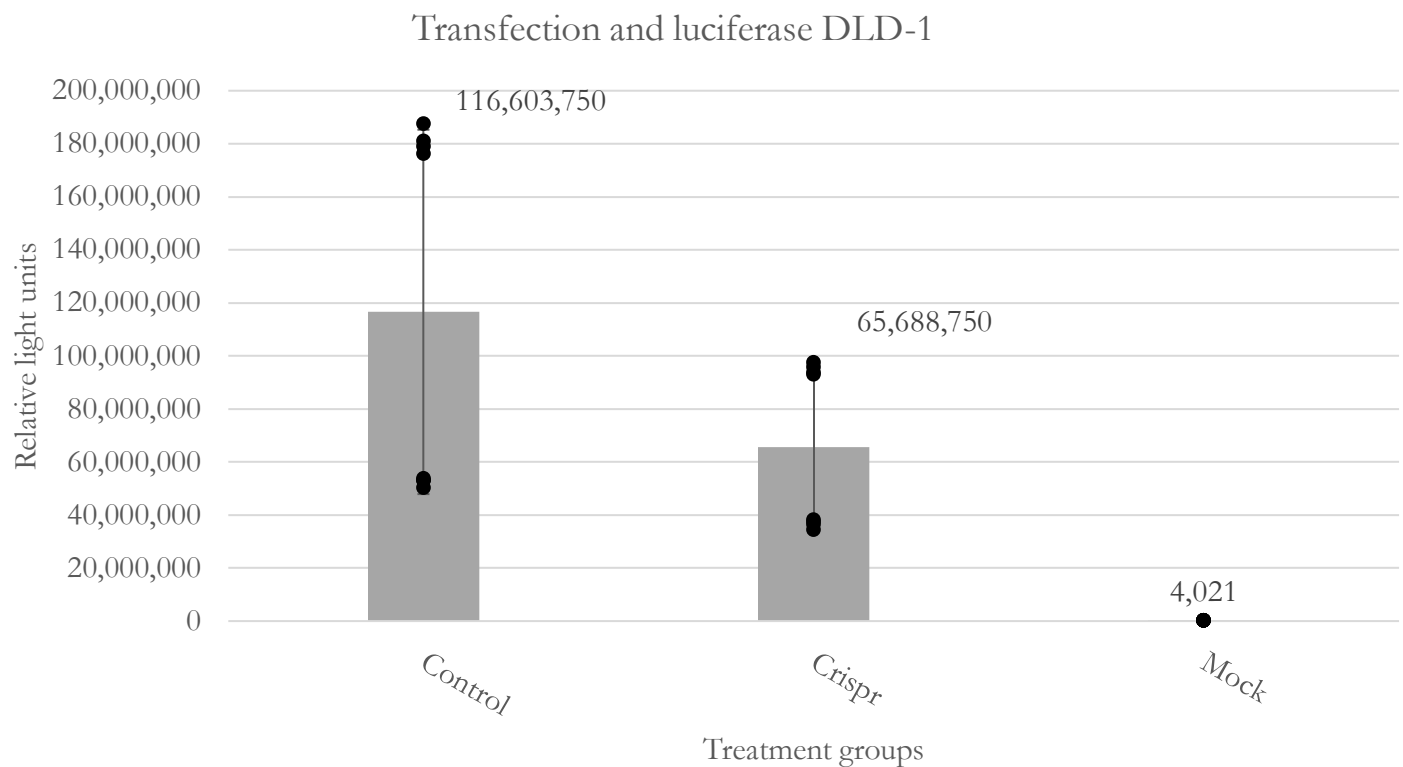


Figure 3. Luciferase activity following transfection of DLD-1. Cells were transfected with pUC18 (Mock), pSelect-Zeo-Lucia-AAVS1 (control), or pSelect-Zeo-Lucia-AAVS1 together with AAVS1-T2-CRISPR (CRISPR). Luciferase activity was measured in culture supernatants using the QUANTI-Luc assay and expressed as relative light units (RLU). Bars represent mean $\pm$ SD. Made with excel.

Comparison between DLD-1 and Caco-2 Transfection and Luciferase assay

DLD-1 cells demonstrated strong luciferase activity in the reporter-containing groups, whereas mock-transfected cells showed only minimal background signal.

Luciferase expression was generally higher in DLD-1 cells in figure 3 compared to the Caco-2 cells in figure 2, suggesting differences in transfection efficiency, plasmid uptake, or cellular expression capacity between the two cell lines.

Results from Luciferase proliferation assay

Figure 4 below shows the luciferase proliferation assay of the caco-2 cells. The luciferase activity remained detectable throughout the proliferation assay in both treatment groups. The CRISPR-transfected cells maintained luciferase expression over the experimental period of 4 days, consistent with stable genomic integration of the reporter construct. The control group also displayed measurable luciferase activity; however, transiently transfected plasmids are expected to become diluted during cell division, resulting in reduced reporter expression over time.

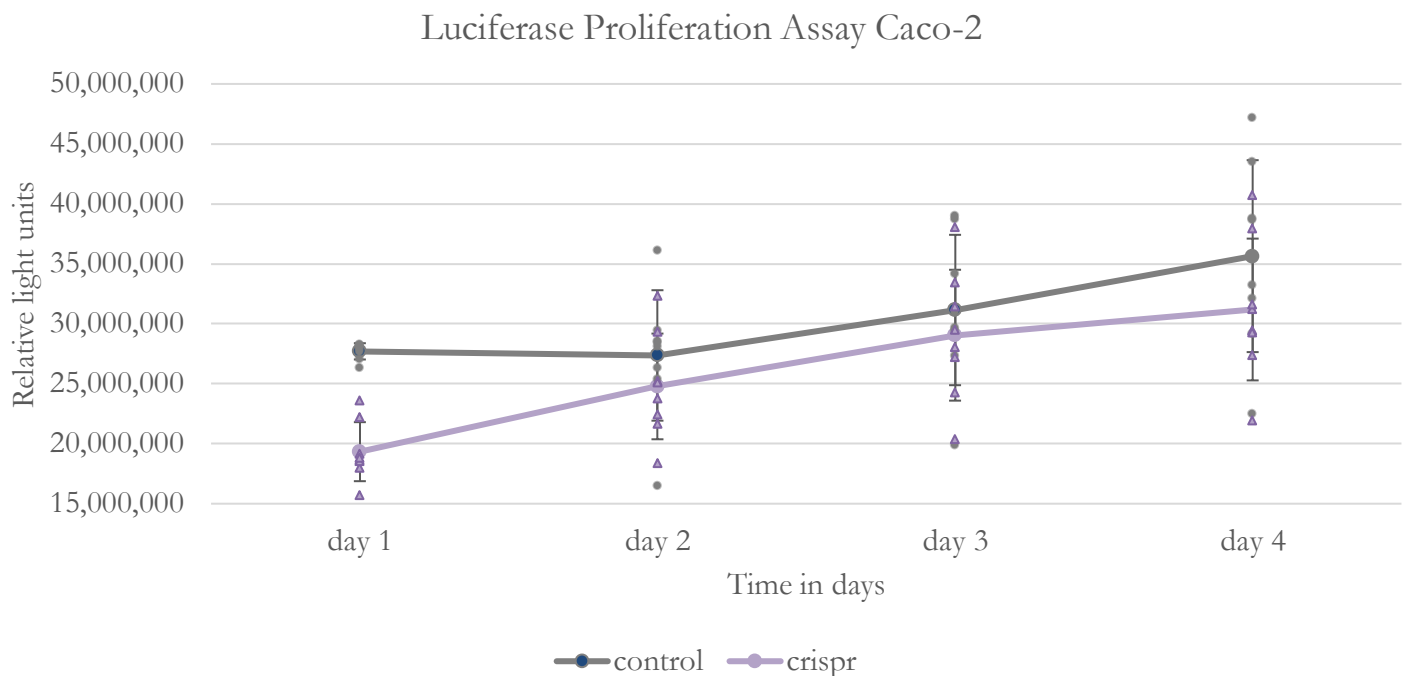


Figure 4. Luciferase activity in proliferating Caco-2 cells at Day 1, 2, 3 and, 4 following transfection. Caco-2 cells transfected with either the Lucia reporter plasmid alone (Control) or the Lucia reporter plasmid together with AAVS1-T2-CRISPR/Cas9 (CRISPR) were monitored during proliferation. Luciferase activity was measured in culture supernatants on Day 1 and Day 4 using the QUANTI-Luc assay and expressed as relative light units (RLU). Bars represent mean $\pm$ SD of eight technical replicates. Made with excel.

Discussion:

The purpose of the Transition and Luciferase assay was to evaluate luciferase expression following transient transfection and CRISPR-mediated genomic integration of a Lucia reporter construct. Successful transfection was demonstrated by the markedly increased luciferase activity observed in the control and CRISPR-treated samples compared with the mock-transfected cells, in both caco-2 cell line and DLD-1 cell line, for the Transition and Luciferase assay.

The mock-transfected samples exhibited only low luminescence values, indicating minimal background signal from the assay system. This confirms that the observed luminescence in the reporter-containing groups originated from expression of the Lucia reporter gene.

Both Caco-2 and DLD-1 cells were successfully transfected; however, variation was observed between replicates and between cell lines.

Transfection was successfully carried out, and the cells remained in generally good condition through the procedure. After transfection, the cells were sufficiently healthy and confluent for the luciferase assay. Signs of severe morphological changes or extensive cell death appeared minimal when checking the cells, and the proliferation assay could continue.

The purpose of the proliferation assay was to evaluate the stability of luciferase expression over time. Since the CRISPR construct targets the AAVS1 safe-harbor locus, genomic integration should allow stable inheritance of the reporter gene during cell division. In contrast, reporter expression from transient plasmid transfection is expected to decrease as plasmids become diluted during proliferation. Although the experimental period was relatively short, the results were consistent with continued reporter expression in the CRISPR-treated cells.

A limitation of the experiment was the low number of replicates, which limits statistical power and increases uncertainty in the calculated standard deviations. Additional biological replicates would improve the reliability of the conclusions.

Variation occur across all the samples; however, this is to be expected in transfection-based reporter assay as there are a lot of variables when transfecting such as efficiency, cell number, plasmid uptake and DNA distribution between cells. Luciferase readout can also be sensitive to small changes in cell state.

Conclusion:

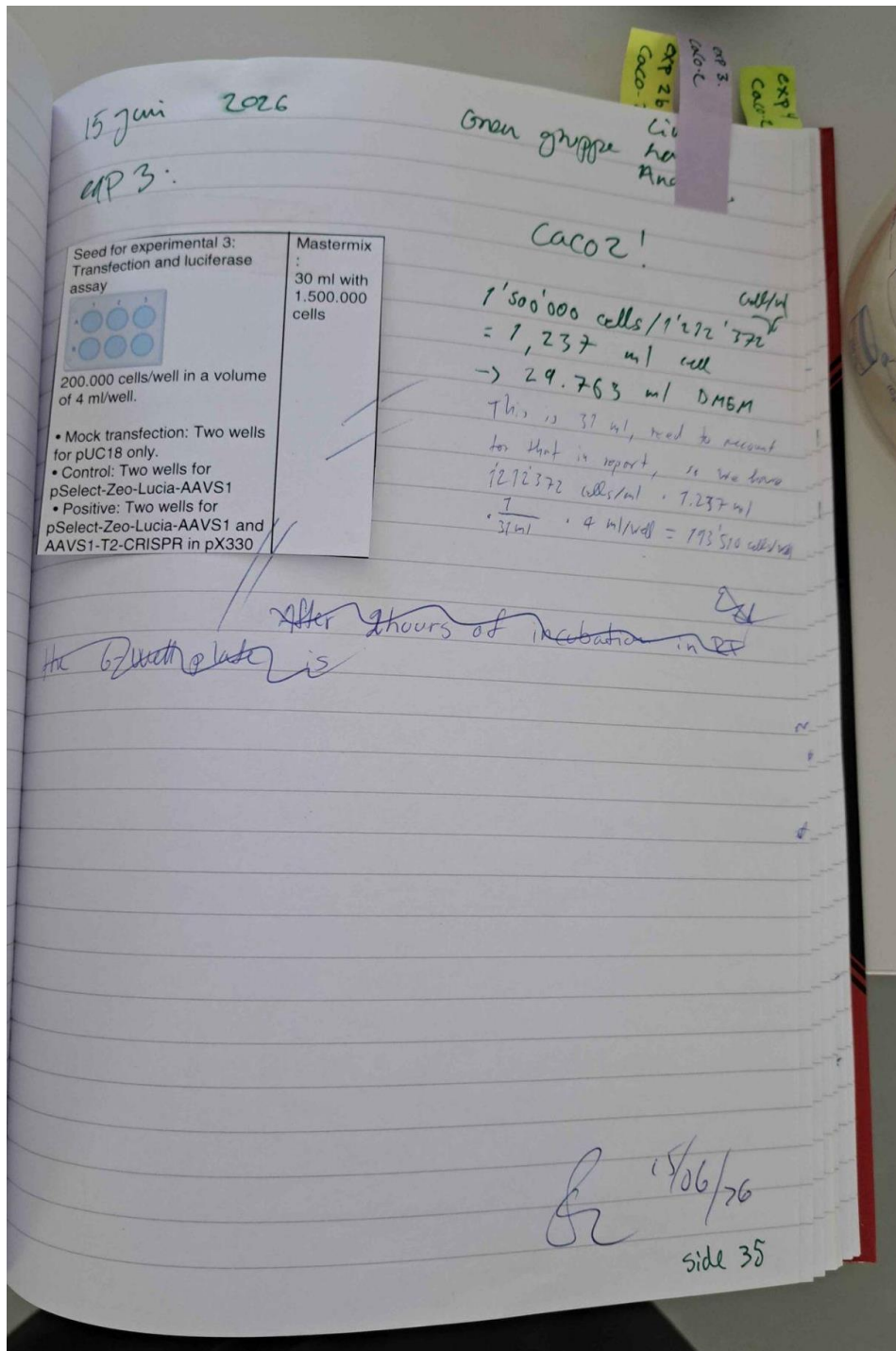
The experiment successfully demonstrated luciferase reporter expression following transfection of Caco-2 and DLD-1 cells. Both the control and CRISPR-transfected groups showed substantially higher luciferase activity than mock-transfected cells, confirming successful delivery and expression of the reporter construct. Furthermore, the proliferation assay indicated luciferase expression in CRISPR-treated cells and control cells. Due to the timeline of the experiment we could not demonstrate the stable chromosomal integration of the Lucia construct.

Appendix:

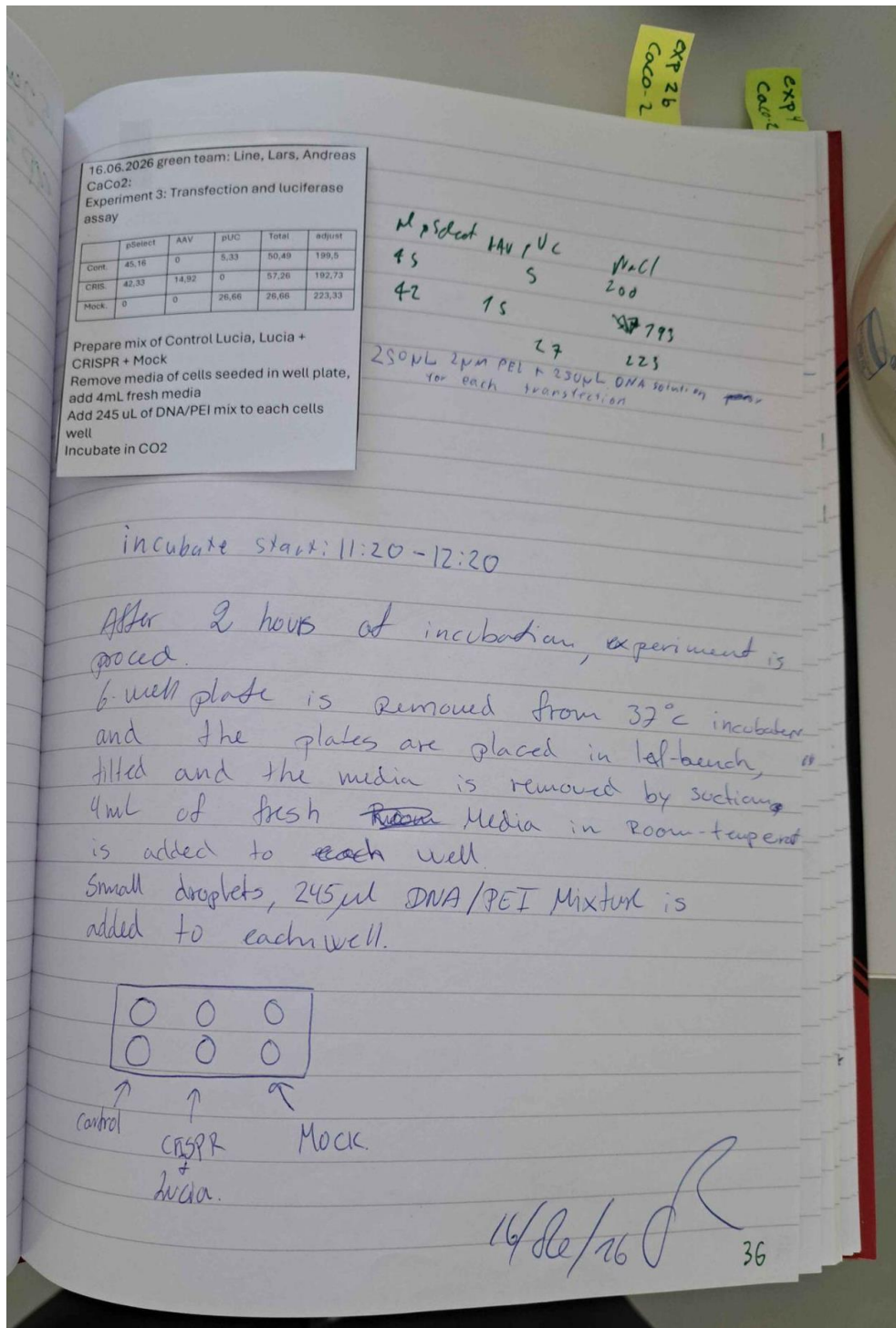
Materials

1. Caco-2 cells (week 1)
2. DLD-1 cells (week 2)
3. 6 well Nunclon Delta treated cell culture plates
4. 85 mM sterile Sodium citrate (SC), sterile filtered.
5. PBS w/o Ca and Mg (PBSw/o)
6. 150 mM NaCl, sterile filtered.
7. Growth medium, complete: DMEM supplemented with 10% Fetal Bovine Serum (FBS), and 100 U/mL Penicillin-Streptomycin (P/S)
8. 2 μ M Polyethyleneimine (PEI) (Alfa Aesar)
9. Plasmids for promoter reporter assays: a. AAVS1-T2-CRISPR in pX330 b. pSelect-Zeo-Lucia-AAVS1 c. pUC18: Plasmid without eukaryotic gene regulatory sequence domains.
10. Phosphate Buffered Saline with Ca^{2+} and Mg^{2+} (PBSw)
11. Quanti-luc
12. GloMax-96 Microplate Luminometer with dual injectors (Promega).
13. White 96-well plates with flat bottom (Costar).
14. 70% ethanol.

Monday:



On the first day, cells trypsinized in the first experiment was used to seed a 6 well plate with 200.000 cells each. These 6 wells consist of two we labelled mock, containing only pUC18. Two wells labelled control consisting of pSelect-Zeo-Lucia-AAVS1 and two wells labelled positive (C+L), containing pSelect-Zeo-Lucia-AAVS1 and AAVS1-T2-CRISPR in PX330. The wells were filled with DIEM media and incubated over night.



Tuesday:

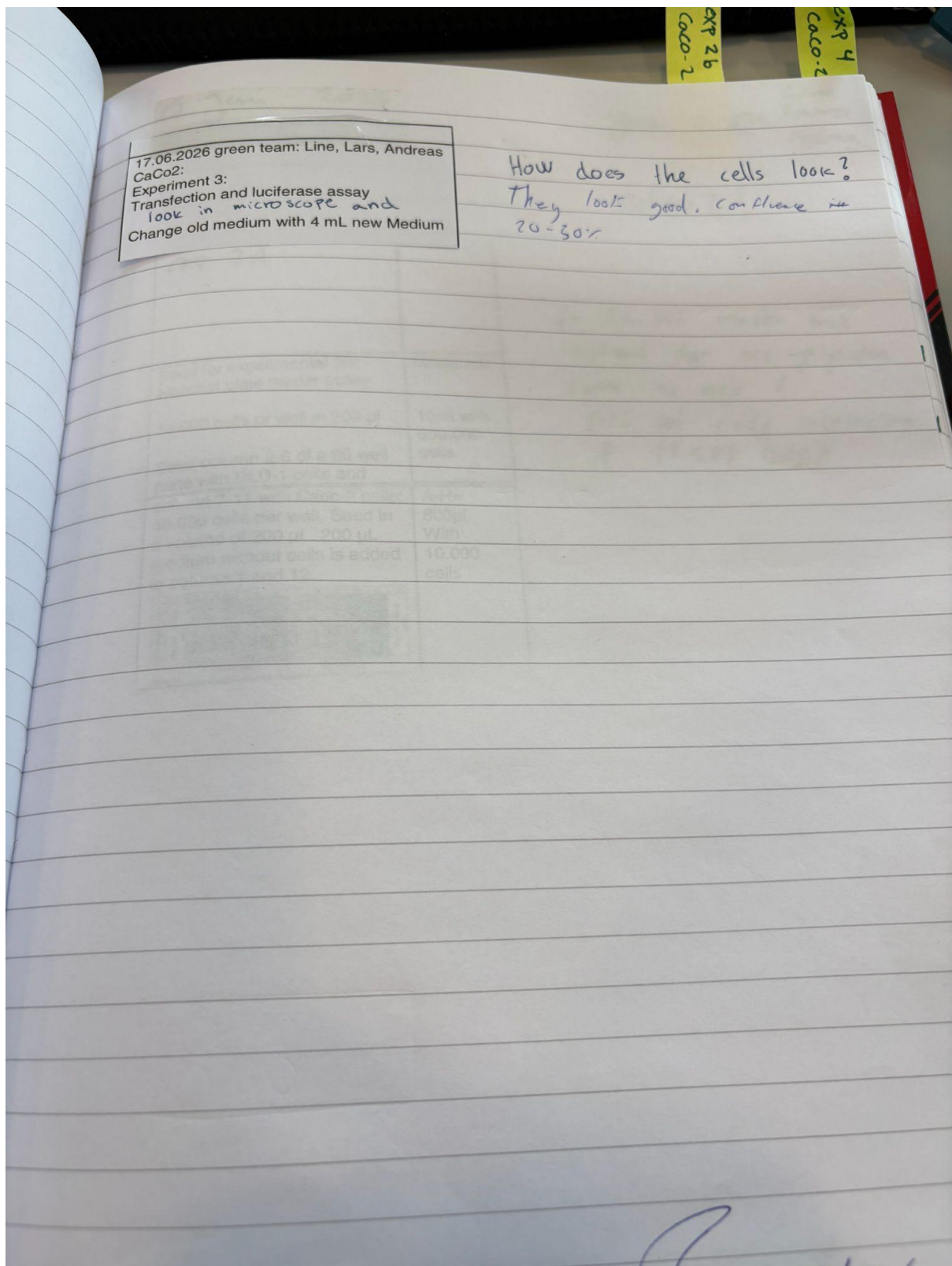
Second day, preparations of mixes of the Control, Mock and Positives were made with DNA solutions. Calculations were done to ensure that the mixes had a total of 250 uL in total. Each DNA solution was mixed with 250 uL 2uM PEI.

	pSelect	AAV	pUC	Total	adjust
Control	45,16	0	5,33	50,49	199,5
CRISPR	42,33	14,92	0	57,26	192,73
Mock	0	0	26,66	26,66	223,33

Table 1: Table shows the transfection mixes used for the different conditions. The volumes are given in μL .

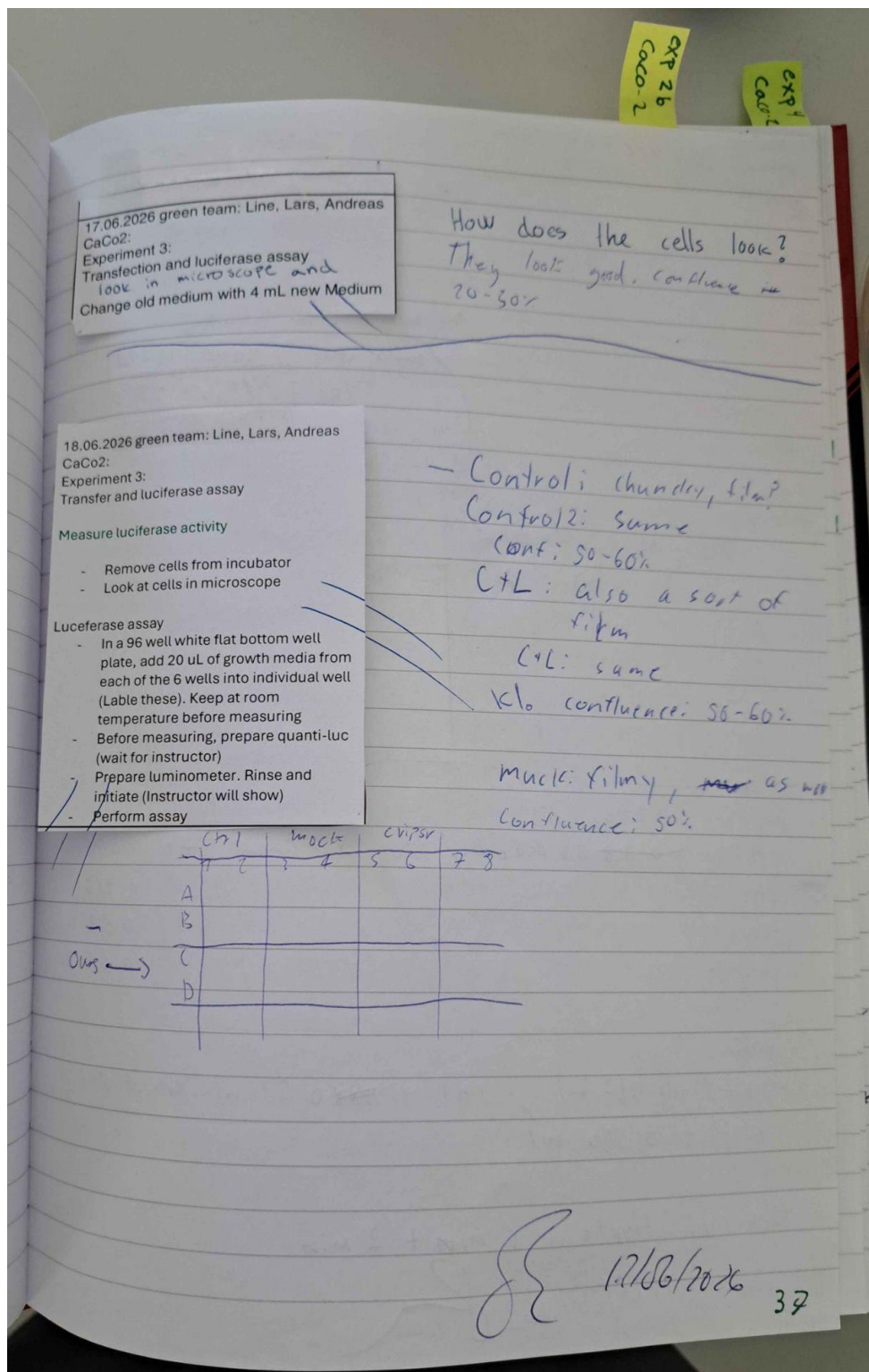
Once mixes were done, the old media was removed and 4 mL new media was added to the cells seeded in the plates and put to incubation. After incubation of 2 hours, the well plates are removed from the incubator and media is removed by suction. Each cell well received 245 μL DNA/PEI mix in form of droplets and thereafter put on incubation for the next 48 hours.

Wednesday:



On the 3rd day, due to the cells needing to incubate for 48 hours, the cells were only checked on to see if they were in good health and to check the confluence of the cells. Agreement was reached that the confluence level was around 20-50% respectively. After that, old medium was removed and the cells were give fresh medium before we continued to let them incubate.

Thursday:



On the fourth day, the 6 well was checked on to see if the cells were in good enough condition for luciferase assay.

Control 1 and 2: Were both in good condition with a Confluence of about 50-60%

Positive (C+L) 1 and 2: Also seemed in good condition with a confluence around 50-60%

Mock 1 and 2: Good health with a confluence of 50% for both.

Due to the good condition of the cells, the luciferase assay could continue. In a 96 well white bottom well plate, 20 uL growth medium from each of the 6 wells were added to 12 of the wells. 2 wells for Control 1, 2 wells for control 2, 2 wells for mock 1 and so on. Once the wells were filled, the white bottom well

plate was kept at room temperature and sealed with parafilm to avoid evaporation before measurement. Our instructor prepared quanti-luc as well as prepared the luminometer and performed the assay. The 6 well plate was put back into incubation.

Friday:

19-06-2026

19.06.2026 green team: Line, Lars, Andreas
 CaCo2:
 Experiment 3:
 Transfection and luciferase assay

Trypsinize the transfected cells.
 Per well:
 Use volumes from table 1.2:
 Wash with 2 ml NaCitrate, use 0.5 ml trypsin
 Count cells

Seed 5000 cells/ml for 1 ml volume in 24 well plate in a column -> need 5 ml in total -> 25'000 cells in total
 Each well (2x crispr and 2x control) in 6 well gets 1 column in 24 well plate. Mock is skipped.

lots of ctrl 100%
 crispr: 85% comp

resuspend in 2.5 ml
 we have 300 for counting

counts: ctrl 1: 2062, 2076
 ctrl 2: 1986, 2040
 crispr 1: 2133, 2100
 crispr 2: 2452, 2960

100 is best
 crispr 2

	ctrl 1	ctrl 2	crispr 1	crispr 2
A	1	2	3	4
B				
C				
D				

200 µl, 0.5 ml (total)
 -> x102 dilution
 concentrations:
 ctrl 1: 320'382 cells/ml
 ctrl 2: 277'076 cells/ml
 crispr 1: 272'644 cells/ml
 crispr 2: 276'710 cells/ml

per col 25'000 cells:
 ctrl 1: ~~320'382~~ 25'000 cells / 320'382 cells/ml -> 0.078
 ctrl 2
 crispr 1
 crispr 2

media:
 ctrl 1: 0.922 ml, ctrl 2: ~~0.885~~ 0.885 ml, crispr 1: 0.910 ml
 crispr 2: 0.970 ml

Obs: Vi trypto i 6 min + 2 min

19/06/26

5Th day we took the 6 well plate and washed it with NaCitrates followed by tryptonization of the transfected cells. Afterwards the cells were counted using a coulter counter in order to seed the cells (5000 cells/ml). After the cells were counted, only the control and C+L cells were seeded in 24 wells, the mock was skipped.

Lab book for week 2:

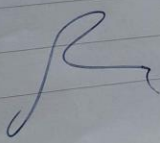
22.06.2026 green team: Line, Lars, Andreas
DLD 1: 7
Experiment 3:

Seed for luciferase assay
200.000 cells/well in a volume of 4 ml/well.
6 wells $\rightarrow$ 1'200'000 cells in 24 ml
Make mastermix of 30 ml with 1'500'000
cells

Cell suspension
1.500.000 cells

1.5 ml cells
Cell suspension cell/ml = ml
1.500.00
 $1.131.180 = 1,326 \text{ ml}$
cell suspension

+ 30ml Media
McCoy

22/06/2026 

72

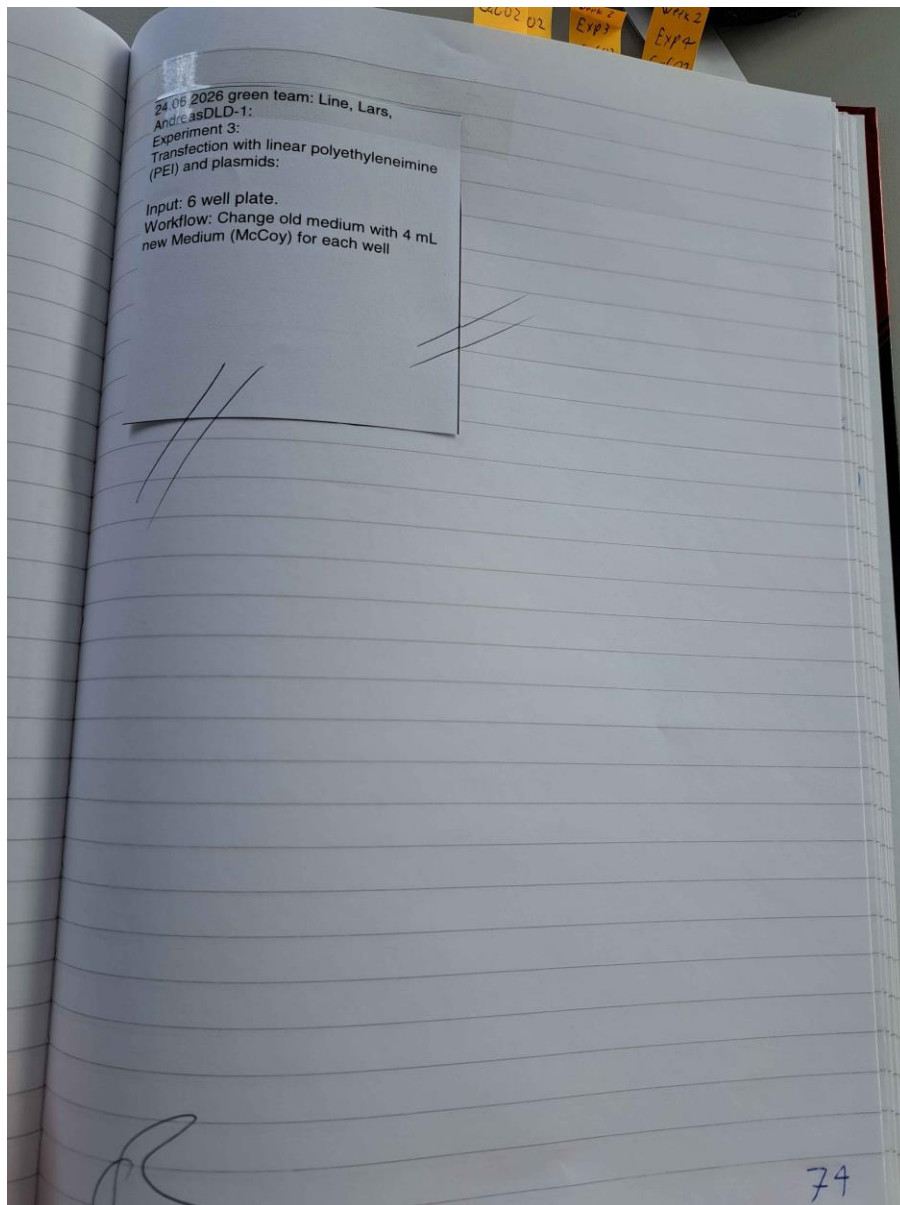
23.06.2026 green team: Line, Lars,
Andreas DLD-1:
Experiment 3:
Transfection with linear polyethylenimine
(PEI) and plasmids

- One hour prior to transfection mix the 250 μ l DNA solution with 250 μ l 2 μ M PEI for each transfection condition.

Transfection mix

	pSelect	AAVS1	pUC18	Total	adjust
Control	45,16 μ l	0	5,333	50,49	199,5
CRISPR	42,338 μ l	14,925	0	57,26	192,73
Mock	0	0	25,66	25,66	223,33

- Vortex briefly and let the reaction stand to form precipitates at room temperature
- Remove media of cells seeded in the 6 well plate and add 4 mL fresh supplemented ~~DMEM~~ ^{McCoy} media to each well.
- Add in small droplets, 245 μ l of DNA / PEI mixture to the cells in each well. Avoid touching plastic surfaces. (PEI precipitates to the wells remember to mix it by pipetting up and down before taking out.)
- Incubate
- Measure luciferase activity on transfected Caco-2 cells (Similar to week 1)



25.06.2026 green team: Line, Lars, Andreas

DLD1:

Experiment 3:

Transfer and luciferase assay

Measure luciferase activity

- Remove cells from incubator
- Look at cells in microscope

Luciferase assay

- In a 96 well white flat bottom well plate, add 20 uL of growth media from each of the 6 wells into individual well (Label these). Keep at room temperature before measuring
- Before measuring, prepare quanti-luc (instructor Presentation)
- Prepare luminometer. Rinse and initiate (instructor presentation)

Perform assay

Handwritten signature

Ctrl: Confluence 40-45%
Grouped in center, ~~lot of dead~~ cells

Ctrl 2: Confluence 90-95%
Grouped in center, lot of dead cells

CRISR: Confluence 80-85%

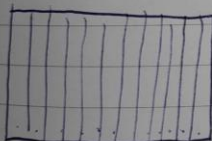
CRISR 2: Confluence 75%

MUCK: Confluence 35-40%
lot of dead cells

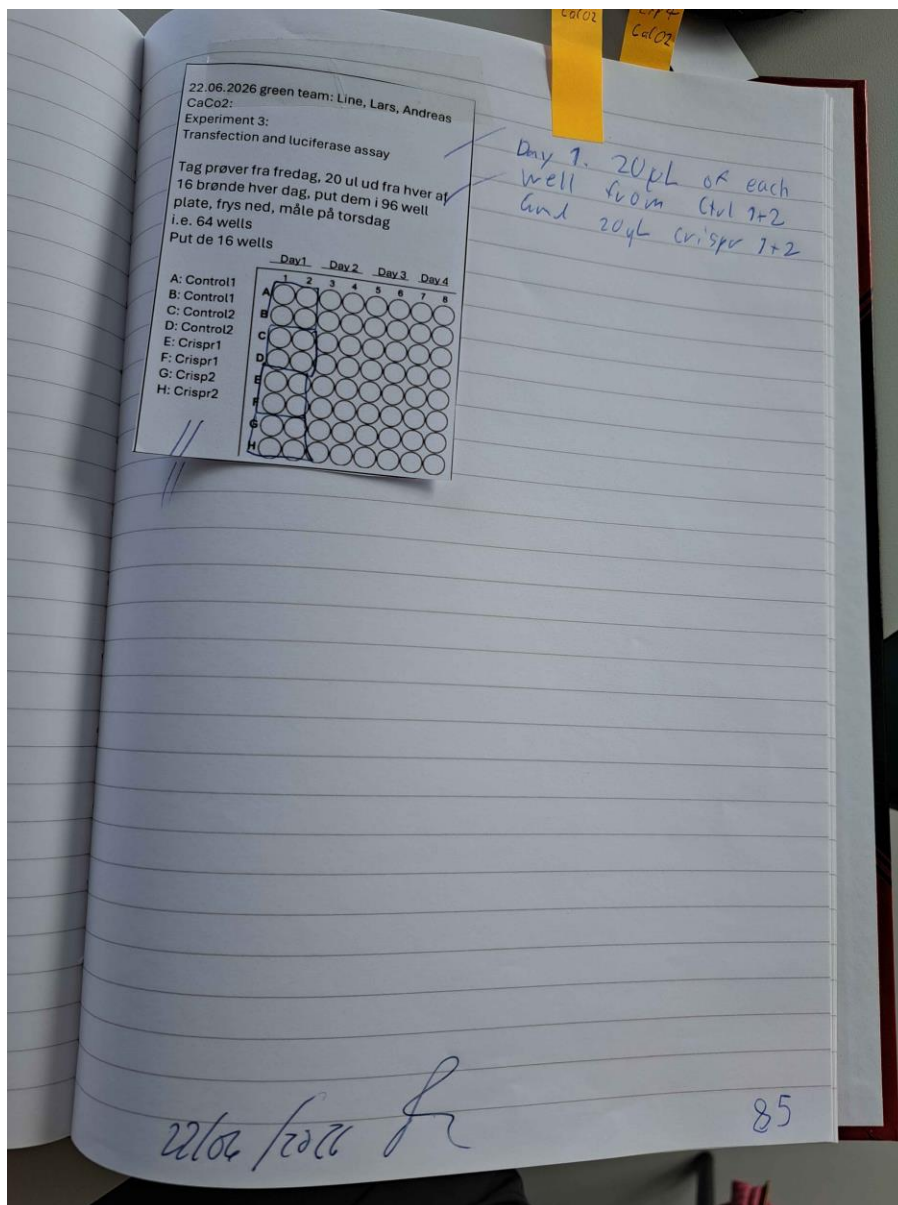
MUCK 2: Confluence 35-40%
lot of dead cells

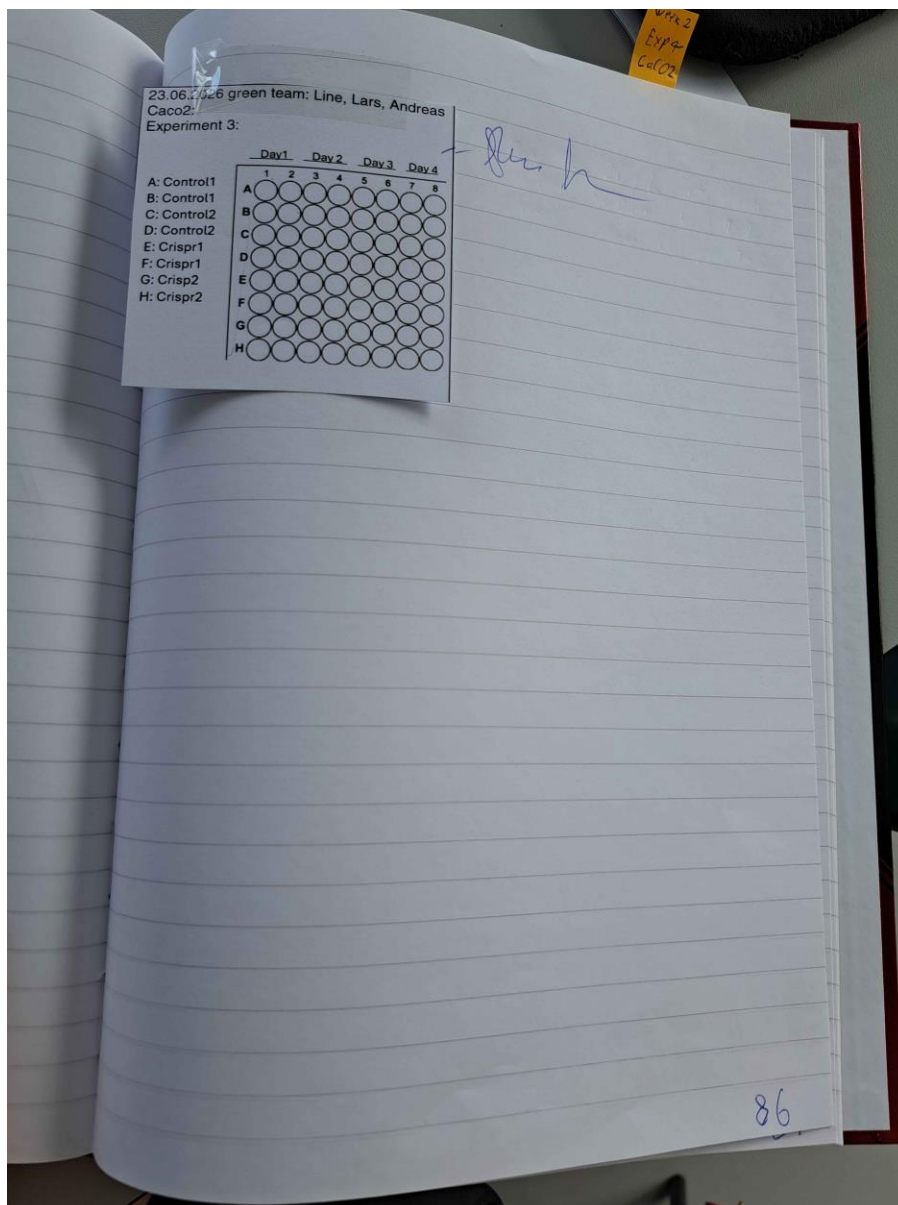
well plating

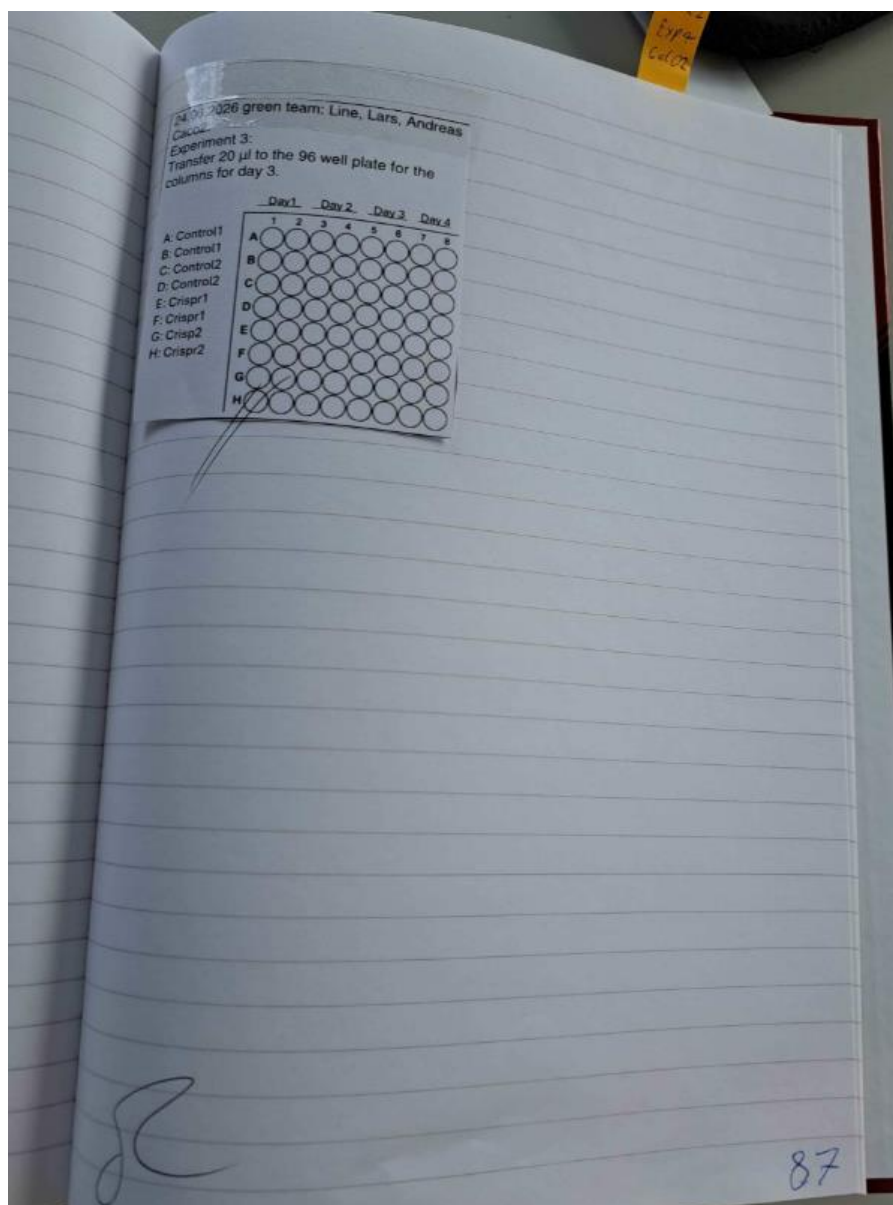
quanti-luc = coelocentrage

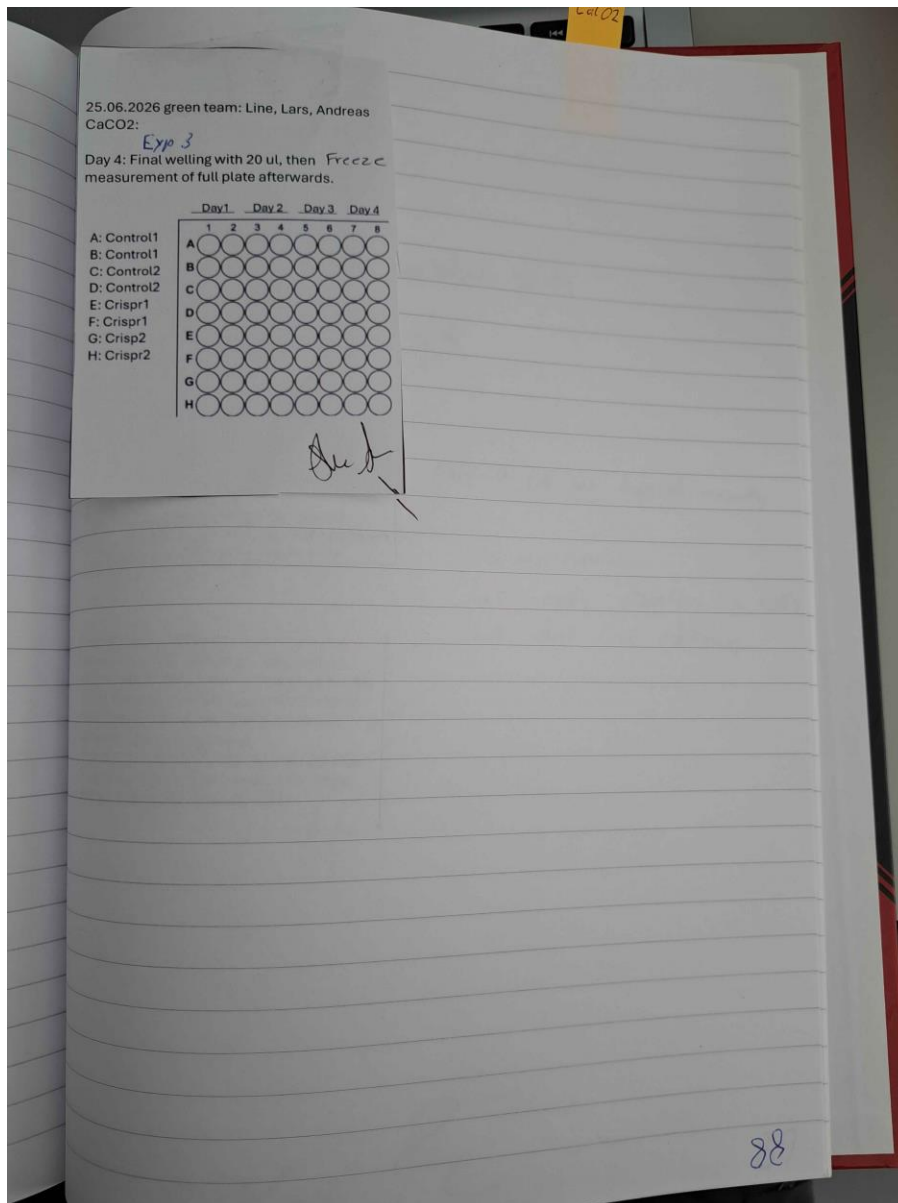


Day 1 2 3 4









24 well layout:

	1	2	3	4	5	6
A	Ctrl 1	Ctrl 2	C+L 1	C+L 2		
B	Ctrl 1	Ctrl 2	C+L 1	C+L 2		
C	Ctrl 1	Ctrl 2	C+L 1	C+L 2		
D	Ctrl 1	Ctrl 2	C+L 1	C+L 2		

Table 2. Layout of the 24 well used for seeding in week 1. Row 5 and 6 were blank and Mock was skipped.

Calculations

15 june calculations:

$1.500.000 \text{ cells} / 1.212.372 \text{ cells/mL} = 1.237 \text{ ml cell} \rightarrow 29.763 \text{ mL DMEM}$

This is 31 ml needed to account so we have $1.212.372 \text{ cells/mL} * 1.237 \text{ mL} * 1/31 \text{ mL} * 4 \text{ mL/well} = 193.510 \text{ cells/well}$

16 june calculations:

DNA calc.

Control: $45,16 \text{ uL/pSelect} + 5,33 \text{ uL/pUC} = 50,49 \text{ uL}$

$250 \text{ uL} - 50,49 \text{ uL} = 199,5 \text{ uL}$

L+C: $42,33 + 14,92 = 57,26$

$250 - 57,26 = 192,73 \text{ uL}$

Mock: $250 \text{ uL} - 26,66 = 223,33 \text{ uL}$

19 june Calculations:

Coulter counter:

Ctrl 1: $3068 + 3214 / 2 = 3141$

Ctrl 2: $1986 + 2240 / 2 = 2113$

C+L: $2178 + 3163 / 2 = 2670$

C+L: $2452 + 2968 / 2 = 2710$

Concentrations:

Ctrl 1: 320.383 cells/mL

Ctrl 2: 278.076 cells/mL

C+L 1: 272.644 cells/mL

C+L 2: 276.420 cells/mL

Per column 25-000 Cells

Ctrl: $25.000 \text{ cells} / 320.383 \text{ cells/mL} = 0.071$

Ctrl 2: $25.000 \text{ cells} / 278.076 \text{ cells/mL} = 0.115$

C+L 1: $25.000 \text{ cells} / 272.644 \text{ cells/mL} = 0.092$

C+L 2: $25.000 \text{ cells} / 276.420 \text{ cells/mL} = 0.090$

Per column Media

Ctrl: 0.922 mL

Ctrl 2: 0.885 mL

C+L: 0.908 mL

C+L 2: 0.910 mL

	Ctrl	Ctrl 2	Crispr	Crispr 2	Muck	Muck 2	Ctrl	Ctrl 2	Crispr	Crispr 2	Muck	Muck 2
	1	2	3	4	5	6	7	8	9	10	11	12
A	5.29E+07	1.79E+08	9.35E+07	3.44E+07	5.10E+03	5.62E+03	5.03E+07	1.81E+08	9.29E+07	3.73E+07	2.52E+03	1.95E+03
B	5.29E+07	1.76E+08	9.58E+07	3.62E+07	6.03E+03	5.93E+03	5.36E+07	1.88E+08	9.74E+07	3.81E+07	2.90E+03	2.11E+03

Figure 5. Raw data of Transfection and luciferase assay of DLD-1 cells

	control 1	control 2	Crispr 1	Crispr 2	Mock 1	Mock 2
	1	2	3	4	5	6
C	29530000.00	28110000.00	24510000.00	48160000.00	6712.00	2530.00
D	37180000.00	28650000.00	24980000.00	48420000.00	10370.00	3250.00

Figure 6. Raw data of Transfection and luciferase assay of DLD-1 cells

None									
0.3		Day 1	Day 1	Day 2	Day 2	Day 3	Day 3	Day 4	Day 4
1		1	2	3	4	5	6	7	8
Control 1	A	2.79E+07	2.71E+07	2.86E+07	2.85E+07	3.87E+07	3.90E+07	4.35E+07	4.72E+07
Control 1	B	2.83E+07	2.63E+07	3.61E+07	2.64E+07	3.42E+07	2.94E+07	3.88E+07	3.87E+07
Control 2	C	2.82E+07	2.76E+07	2.94E+07	2.81E+07	3.10E+07	2.97E+07	3.32E+07	3.21E+07
Control 2	D	2.82E+07	2.80E+07	1.65E+07	2.54E+07	1.99E+07	2.73E+07	2.25E+07	2.92E+07
CRISPR 1	E	2.36E+07	2.22E+07	3.24E+07	2.93E+07	3.81E+07	3.34E+07	4.07E+07	3.80E+07
CRISPR 1	F	1.87E+07	1.80E+07	2.38E+07	2.51E+07	2.95E+07	2.72E+07	3.12E+07	3.16E+07
CRISPR 2	G	1.92E+07	1.86E+07	2.52E+07	2.16E+07	2.81E+07	3.15E+07	2.94E+07	2.93E+07
CRISPR 2	H	1.57E+07	1.88E+07	1.84E+07	2.24E+07	2.04E+07	2.43E+07	2.19E+07	2.74E+07

Figure 5. Raw data of the Transfection and luciferase proliferation assay